

Related Work for Drone Phenology Monitoring

Scope and selection logic

This shortlist is organised around the six themes you specified and ranked within each theme by likely usefulness for a Related Work section on your pipeline: repeated UAV RGB orthomosaics, crown detection with Detectree2/Mask R-CNN, temporal co-registration, graph-based crown association, crown-level RGB phenology, and the joining of field/species labels with future supervision of coarser satellite products. I prioritised 2018–2025 papers, but I kept a small number of older seminal works where they are still the canonical citation for a method or metric. All summaries and relevance notes are grounded in the cited source pages or abstracts. ¹

A few items below are highly relevant but not exact matches to your setting. I flag those explicitly in the “Relation to your work” line so you can decide whether to use them as direct precedent, as adjacent motivation, or as a contrast that your paper resolves. Where metadata were incomplete in the retrieved abstract page, I kept the BibTeX entry minimal rather than guessing fields. ²

UAV-based vegetation and tree monitoring

1. **Simon Ecke et al., 2022, *Remote Sensing*.** “UAV-Based Forest Health Monitoring: A Systematic Review.”

Summary: This review argues that UAVs fill the gap between labour-intensive field plots and coarser traditional remote sensing by enabling flexible, repeated, very-high-resolution forest monitoring workflows.

Relation to your work: Very strong framing citation for the motivation of repeated UAV surveys; your paper is a concrete end-to-end answer to the review’s broader call for scalable tree-level monitoring pipelines. ³

```
@article{ecke2022uavfhm,  
  author = {Ecke, Simon and Dempewolf, Jan and Frey, Julian and others},  
  title = {UAV-Based Forest Health Monitoring: A Systematic Review},  
  journal = {Remote Sensing},  
  year = {2022},  
  volume = {14},  
  number = {13},  
  pages = {3205},  
  doi = {10.3390/rs14133205}  
}
```

1. **Christian Thiel et al., 2020, *Drones*.** “Monitoring Selective Logging in a Pine-Dominated Forest in Central Germany with Repeated Drone Flights Utilizing a Low Cost RTK Quadcopter.”

Summary: A repeated-flight UAV study showing that orthomosaics/CHMs from consumer RTK drones can support individual-tree change detection with low field effort and object-based analysis.

Relation to your work: Strongly relevant as a repeated-flight workflow precedent, but it targets bi-temporal disturbance/change detection rather than temporally stable crown identities and

phenology through long series. Your work extends the idea from two-date change detection to multi-date tree tracking and per-crown phenology. 4

```
@article{thiel2020selectivelogging,  
  author =  
{Thiel, Christian and Mueller, Marlin M. and Berger, Christian and others},  
  title =  
{Monitoring Selective Logging in a Pine-Dominated Forest in Central Germany  
with Repeated Drone Flights Utilizing a Low Cost RTK Quadcopter},  
  journal = {Drones},  
  year = {2020},  
  volume = {4},  
  number = {2},  
  pages = {11},  
  doi = {10.3390/drones4020011}  
}
```

1. **Jae Yong Park et al., 2019, *Remote Sensing*.** “Quantifying Leaf Phenology of Individual Trees and Species in a Tropical Forest Using Unmanned Aerial Vehicle Images.”

Summary: Demonstrates that repeated high-resolution UAV imagery can recover individual-tree and species-level phenology in tropical forest canopies.

Relation to your work: One of the closest ecological precedents for crown-level UAV phenology, but it does not centre the hard computer-vision problem your paper tackles: building temporally stable crown ROIs under drift, illumination change, and crown-shape variability. 5

```
@article{park2019leafphenology,  
  author =  
{Park, Jae Yong and Muller-Landau, Helene C. and Lichstein, Jeremy W. and  
others},  
  title = {Quantifying Leaf Phenology of Individual Trees and Species in a  
Tropical Forest Using Unmanned Aerial Vehicle Images},  
  journal = {Remote Sensing},  
  year = {2019},  
  volume = {11},  
  number = {13},  
  pages = {1534},  
  doi = {10.3390/rs11131534}  
}
```

1. **Nuno Guimarães et al., 2020, *Remote Sensing*.** “Forestry Remote Sensing from Unmanned Aerial Vehicles: A Review Focusing on the Data, Processing and Potentialities.”

Summary: Reviews UAV forestry workflows from acquisition through processing, emphasising the importance of data products such as orthomosaics, point clouds, and canopy models for forest applications.

Relation to your work: Useful broad review citation for your data-processing context; unlike your paper, it does not propose a temporal identity-preserving crown pipeline or phenology labels. 6

```
@article{guimaraes2020forestryuavreview,
  author = {Guimar{\~a}es, Nuno and P{\a}dua, Lu{\'\i}s and Marques, Pedro and others},
  title = {Forestry Remote Sensing from Unmanned Aerial Vehicles: A Review Focusing on the Data, Processing and Potentialities},
  journal = {Remote Sensing},
  year = {2020},
  volume = {12},
  number = {6},
  pages = {1046},
  doi = {10.3390/rs12061046}
}
```

1. **Sabrina Hartling et al., 2021, *GIScience & Remote Sensing*.** "Urban Tree Species Classification Using UAV-Based Multi-Sensor Data Fusion and Deep Learning."

Summary: Shows that UAV data can support urban tree monitoring and species classification when coupled with deep learning and multi-sensor fusion.

Relation to your work: Good urban-tree counterpart to the mostly forest-oriented literature; your study differs by making temporal stability of individual crowns the central problem, rather than one-off classification from a single acquisition. 7

```
@article{hartling2021urbantreespecies,
  author = {Hartling, Sabrina and Sagan, Vedrana and Sidike, Paheding and others},
  title = {Urban Tree Species Classification Using UAV-Based Multi-Sensor Data Fusion and Deep Learning},
  journal = {GIScience \& Remote Sensing},
  year = {2021},
  volume = {58},
  number = {8},
  pages = {1274--1295}
}
```

Individual tree crown detection and delineation

1. **James G. C. Ball et al., 2023, *Remote Sensing in Ecology and Conservation*.** "Accurate Delineation of Individual Tree Crowns in Tropical Forests from Aerial RGB Imagery Using Detectree2."

Summary: Introduces Detectree2, adapting Mask R-CNN to delineate irregular tree-crown polygons from aerial RGB imagery in challenging dense tropical forests.

Relation to your work: This is the most direct detector citation for your pipeline because you use Detectree2. Your contribution begins where theirs ends: stabilising those date-wise detections into temporal chains and consensus crowns for phenology extraction. 8

```
@article{ball2023detectree2,
  author = {Ball, James G. C. and Aubry-Kientz, M{\e}laine and Vincent, Gr{\e}goire and others},
  title = {Accurate Delineation of Individual Tree Crowns in Tropical Forests from
```

```
Aerial RGB Imagery Using Detectree2},
  journal = {Remote Sensing in Ecology and Conservation},
  year = {2023},
  doi = {10.1002/rse2.332}
}
```

1. **Ben G. Weinstein et al., 2020, *Methods in Ecology and Evolution*.** "DeepForest: A Python Package for RGB Deep Learning Tree Crown Delineation."

Summary: Presents DeepForest as an easy-to-use RGB tree-crown detection package with a large pre-trained model and fine-tuning workflow.

Relation to your work: Important contrast citation: DeepForest popularised RGB crown detection, but it outputs boxes rather than crown masks; your paper can explicitly argue that polygon masks are better than boxes for crown-pure phenology ROIs. ⁹

```
@article{weinstein2020deepforest,
  author = {Weinstein, Ben G. and Marconi, Sergio and Aubry-Kientz, M{\`e}laine and others},
  title = {DeepForest: A Python Package for RGB Deep Learning Tree Crown Delineation},
  journal = {Methods in Ecology and Evolution},
  year = {2020},
  volume = {11},
  number = {12},
  pages = {1743--1751},
  doi = {10.1111/2041-210X.13472}
}
```

1. **Ben G. Weinstein et al., 2019, *Remote Sensing*.** "Individual Tree-Crown Detection in RGB Imagery Using Semi-Supervised Deep Learning Neural Networks."

Summary: Shows that RGB crown detection can be bootstrapped with noisy LiDAR-derived labels and then refined with hand annotations, establishing a strong RGB-only detection baseline.

Relation to your work: Good citation for the broader move from hand-crafted crown delineation to learned RGB detectors; however, this work remains detection-focused and does not address temporal linking or phenology ROI stability. ¹⁰

```
@article{weinstein2019semisupervisedcrowns,
  author = {Weinstein, Ben G. and Marconi, Sergio and Bohlman, Stephanie and Zare, Alina and White, Ethan},
  title = {Individual Tree-Crown Detection in RGB Imagery Using Semi-Supervised Deep Learning Neural Networks},
  journal = {Remote Sensing},
  year = {2019},
  volume = {11},
  number = {11},
  pages = {1309},
}
```

```
doi = {10.3390/rs11111309}
}
```

1. **Kaiming He et al., 2017, *ICCV*. "Mask R-CNN."**

Summary: Canonical instance-segmentation architecture that extends Faster R-CNN with a parallel pixel-mask prediction branch.

Relation to your work: Core methodological citation for why crown masks, not only boxes, are available and useful. In your paper, this supports the claim that polygonal crown support is preferable for phenology features and crown-consensus geometry. ¹¹

```
@inproceedings{he2017maskrcnn,
  author = {He, Kaiming and Gkioxari, Georgia and Doll{\`a}r, Piotr and
Girshick, Ross},
  title = {Mask R-CNN},
  booktitle = {Proceedings of the IEEE International Conference on Computer
Vision},
  year = {2017},
  pages = {2961--2969},
  doi = {10.1109/ICCV.2017.322}
}
```

1. **Kunyong Yu et al., 2022, *Remote Sensing*. "Comparison of Classical Methods and Mask R-CNN for Automatic Tree Detection and Mapping Using UAV Imagery."**

Summary: Directly compares local maxima, marker-controlled watershed, and Mask R-CNN, finding the deep model stronger overall and documenting tile-based training/inference plus IoU-based crown evaluation.

Relation to your work: Useful comparator for arguing why deep instance segmentation is the right front-end for crown extraction; they stop at single-date crown detection, whereas your contribution is the multi-date association and consensus-ROI stage. ¹²

```
@article{yu2022classicalvsmaskrcnn,
  author = {Yu, Kunyong and Hao, Zhixin and Post, Christopher J. and others},
  title = {Comparison of Classical Methods and Mask R-CNN for Automatic Tree
Detection and Mapping Using UAV Imagery},
  journal = {Remote Sensing},
  year = {2022},
  volume = {14},
  number = {2},
  pages = {295},
  doi = {10.3390/rs14020295}
}
```

1. **Ben G. Weinstein et al., 2021, *PLOS Computational Biology*. "A Benchmark Dataset for Canopy Crown Detection and Delineation in Co-Registered Airborne RGB, LiDAR and Hyperspectral Imagery from the National Ecological Observatory Network."**

Summary: Provides a public benchmark and evaluation framework for crown detection/delineation methods across forest types.

Relation to your work: Excellent benchmarking citation if you want to situate Detectree2 and

crown-delineation choices within the broader evaluation ecosystem; it does not address temporal stability, which is precisely your gap. ¹³

```
@article{weinstein2021crownbenchmark,  
  author = {Weinstein, Ben G. and Marconi, Sergio and Bohlman, Stephanie and  
    others},  
  title = {A Benchmark Dataset for Canopy Crown Detection and Delineation in  
    Co-Registered Airborne RGB, LiDAR and Hyperspectral Imagery from the National  
    Ecological Observatory Network},  
  journal = {PLOS Computational Biology},  
  year = {2021},  
  volume = {17},  
  number = {6},  
  pages = {e1009180},  
  doi = {10.1371/journal.pcbi.1009180}  
}
```

Multi-temporal orthomosaic alignment and co-registration

1. Manuel Guizar-Sicairos, Samuel T. Thurman, and James R. Fienup, 2008, *Optics Letters*.

“Efficient Subpixel Image Registration Algorithms.”

Summary: The classic subpixel phase-correlation paper; it provides fast, accurate translation registration in the Fourier domain.

Relation to your work: This is the canonical citation for your use of phase cross-correlation; your tiled/windowed PCC is an application-specific adaptation for orthomosaic drift rather than a new registration primitive. ¹⁴

```
@article{guizar2008subpixel,  
  author = {Guizar-Sicairos, Manuel and Thurman, Samuel T. and Fienup, James  
    R.},  
  title = {Efficient Subpixel Image Registration Algorithms},  
  journal = {Optics Letters},  
  year = {2008},  
  volume = {33},  
  number = {2},  
  pages = {156--158},  
  doi = {10.1364/OL.33.000156}  
}
```

1. Irene Aicardi et al., 2016, *Remote Sensing*. “An Image-Based Approach for the Co-Registration of Multi-Temporal UAV Image Datasets.”

Summary: Shows that multi-temporal UAV datasets can be co-registered without ground control points by constraining later epochs to a reference image block.

Relation to your work: A central precedent for the claim that consumer-drone photogrammetry still leaves relative alignment as an open issue; your method differs by applying lightweight image-domain alignment to finished orthomosaics and then propagating shifts to crown geometries. ¹⁵

```
@article{aicardi2016multitemporaluavcoreg,
  author = {Aicardi, Irene and Nex, Francesco and Gerke, Markus and Lingua, Andrea Maria},
  title = {An Image-Based Approach for the Co-Registration of Multi-Temporal UAV Image Datasets},
  journal = {Remote Sensing},
  year = {2016},
  volume = {8},
  number = {9},
  pages = {779},
  doi = {10.3390/rs8090779}
}
```

1. **E. W. Nota et al., 2022, *Earth Surface Processes and Landforms*.** “Improving UAV-SfM Time-Series Accuracy by Co-Alignment and Contributions of Ground Control or RTK Positioning.”
Summary: Empirically shows that co-alignment substantially improves relative accuracy in UAV-SfM time series and can reduce relative errors to below 2 cm.
Relation to your work: Very useful for motivating why even RTK/GNSS-based workflows still need temporal alignment attention; your paper addresses the orthomosaic/crown domain rather than 3D DSM comparison. ¹⁶

```
@article{nota2022coalignment,
  author = {Nota, E. W. and others},
  title = {Improving UAV-SfM Time-Series Accuracy by Co-Alignment and Contributions of Ground Control or RTK Positioning},
  journal = {Earth Surface Processes and Landforms},
  year = {2022}
}
```

1. **Tjalling de Haas et al., 2021, *Frontiers in Remote Sensing*.** “Optimization of Topographic Change Detection With UAV Structure-From-Motion Photogrammetry Through Survey Co-Alignment.”
Summary: Compares classical per-survey processing with co-alignment and shows that co-alignment improves relative accuracy for change detection by roughly a factor of three to four.
Relation to your work: Strong support for the principle that temporal consistency matters at least as much as absolute geolocation when comparing repeated UAV products; your pipeline makes the same point for crown identities rather than terrain surfaces. ¹⁷

```
@article{dehaas2021surveycoalignment,
  author = {de Haas, T. and Nijland, W. N. and McArdell, B. W. and Kalthof, M. W. M. L.},
  title = {Optimization of Topographic Change Detection With UAV Structure-From-Motion Photogrammetry Through Survey Co-Alignment},
  journal = {Frontiers in Remote Sensing},
  year = {2021},
  volume = {2},
  pages = {626810},
}
```

```
doi = {10.3389/frsen.2021.626810}
}
```

1. **Taeheon Kim et al., 2022, *Geocarto International*.** “Multi-Temporal Orthophoto and Digital Surface Model Registration Produced from UAV Imagery Over an Agricultural Field.”

Summary: Proposes an explicit registration workflow for multi-temporal UAV orthophotos and DSMs, addressing geometric inconsistencies without relying solely on field-survey GCP maintenance.

Relation to your work: Useful adjacent citation because it is orthophoto-focused rather than point-cloud-focused; however, it is agricultural and not tree-centric, so your work addresses a harder object-level temporal association problem in cluttered canopy scenes. 18

```
@article{kim2022orthophotodsmregistration,
  author =
    {Kim, Taeheon and Park, Jueon and Lee, Changhui and Yun, Yerin and Jung,
    Jinha and Han, Youkyung},
  title = {Multi-Temporal Orthophoto and Digital Surface Model Registration
    Produced from UAV Imagery Over an Agricultural Field},
  journal = {Geocarto International},
  year = {2022},
  volume = {37},
  number = {27},
  pages = {18767--18790},
  doi = {10.1080/10106049.2022.2143913}
}
```

1. **Georgios D. Evangelidis and Emmanouil Z. Psarakis, 2008, *IEEE TPAMI*; Ethan Rublee et al., 2011, *ICCV*.** “Parametric Image Alignment Using Enhanced Correlation Coefficient Maximization” and “ORB: An Efficient Alternative to SIFT or SURF.”

Summary: ECC is the classic direct-alignment baseline for parametric registration, whereas ORB is a standard fast feature-matching baseline.

Relation to your work: These are ideal “why we did not use only feature matching / global parametric alignment” citations; your tiled PCC is a simpler, translation-focused alternative that better matches residual local drift between already-generated orthomosaics. 19

```
@article{evangelidis2008ecc,
  author = {Evangelidis, Georgios D. and Psarakis, Emmanouil Z.},
  title = {Parametric Image Alignment Using Enhanced Correlation Coefficient
    Maximization},
  journal = {IEEE Transactions on Pattern Analysis and Machine Intelligence},
  year = {2008},
  volume = {30},
  number = {10},
  pages = {1858--1865}
}
```

```
@inproceedings{rublee2011orb,
  author =
    {Rublee, Ethan and Rabaud, Vincent and Konolige, Kurt and Bradski, Gary},
  title = {ORB: An Efficient Alternative to SIFT or SURF},
```

```

booktitle = {Proceedings of the IEEE International Conference on Computer
Vision},
year = {2011},
pages = {2564--2571},
doi = {10.1109/ICCV.2011.6126544}
}

```

Object association and tracking-by-detection

1. **Despoina Touska et al., 2023, *MMM*.** "Graph-Based Data Association in Multiple Object Tracking: A Survey."

Summary: Surveys graph-based formulations of data association in multi-object tracking, explicitly framing trajectories as graph structures over discrete detections.

Relation to your work: This is the cleanest high-level citation for your graph-based temporal association framing. It helps justify why a graph of per-date crowns is more expressive than one-shot frame-to-frame assignment. 20

```

@inproceedings{touska2023graphsurvey,
  author = {Touska, Despoina and Gkountakos, Konstantinos and Tsikrika,
Theodora and Ioannidis, Konstantinos and Vrochidis, Stefanos and
Kompatsiaris, Ioannis},
  title = {Graph-Based Data Association in Multiple Object Tracking: A
Survey},
  booktitle = {Proceedings of MultiMedia Modeling},
  year = {2023},
  pages = {386--398},
  doi = {10.1007/978-3-031-27818-1_32}
}

```

1. **Harold W. Kuhn, 1955, *Naval Research Logistics Quarterly*.** "The Hungarian Method for the Assignment Problem."

Summary: Classic bipartite assignment algorithm used throughout tracking-by-detection literature for one-to-one frame-to-frame matching.

Relation to your work: Essential baseline citation for strict one-to-one matching. Your paper can explicitly argue that crown splits, merges, containment, and missed detections violate the assumptions under which plain Hungarian matching is most natural. 21

```

@article{kuhn1955hungarian,
  author = {Kuhn, Harold W.},
  title = {The Hungarian Method for the Assignment Problem},
  journal = {Naval Research Logistics Quarterly},
  year = {1955},
  volume = {2},
  number = {1-2},
  pages = {83--97},
}

```

```
doi = {10.1002/nav.3800020109}
}
```

1. Alex Bewley et al., 2016, *ICIP*. "Simple Online and Realtime Tracking."

Summary: A canonical tracking-by-detection baseline combining Kalman filtering with Hungarian assignment, notable for its simplicity and reliance on detection quality.

Relation to your work: Very helpful as the "strict local assignment" baseline that your graph formulation departs from. SORT's local one-step association is exactly the kind of formulation that becomes brittle when crowns are missed or reshaped between dates. ²²

```
@inproceedings{bewley2016sort,
  author = {Bewley, Alex and Ge, Zongyuan and Ott, Lionel and Ramos, Fabio
and Upcroft, Ben},
  title = {Simple Online and Realtime Tracking},
  booktitle = {Proceedings of the IEEE International Conference on Image
Processing},
  year = {2016},
  pages = {3464--3468},
  doi = {10.1109/ICIP.2016.7533003}
}
```

1. Nicolai Wojke, Alex Bewley, and Dietrich Paulus, 2017, *ICIP*. "Simple Online and Realtime Tracking with a Deep Association Metric."

Summary: Extends SORT with appearance embeddings to reduce identity switches through occlusions and short-term ambiguity.

Relation to your work: Useful analogue for adding richer similarity cues beyond geometry alone. Your crown-linking score plays a comparable role, but with remote-sensing-specific overlap, centroid, and shape cues instead of person ReID features. ²³

```
@inproceedings{wojke2017deepsort,
  author = {Wojke, Nicolai and Bewley, Alex and Paulus, Dietrich},
  title = {Simple Online and Realtime Tracking with a Deep Association
Metric},
  booktitle = {Proceedings of the IEEE International Conference on Image
Processing},
  year = {2017},
  pages = {3645--3649},
  doi = {10.1109/ICIP.2017.8296962}
}
```

1. Xiaolong Jiang et al., 2019, *arXiv*. "Graph Neural Based End-to-End Data Association Framework for Online Multiple-Object Tracking."

Summary: Formulates frame-by-frame association as a graph problem and learns affinities over graph edges instead of relying entirely on hand-built costs.

Relation to your work: Good modern graph-association citation. Your method is more interpretable and ecology-specific: you use hand-crafted crown similarity features and explicit ecological edge cases instead of end-to-end learned association. ²⁴

```
@article{jiang2019gnndataassociation,
  author = {Jiang, Xiaolong and Li, Peizhao and Li, Yanjing and Zhen, Xiantong},
  title = {Graph Neural Based End-to-end Data Association Framework for Online Multiple-Object Tracking},
  journal = {arXiv preprint arXiv:1907.05315},
  year = {2019}
}
```

1. **Yihong Wang et al., 2020, ICRA.** "Joint Object Detection and Multi-Object Tracking with Graph Neural Networks."

Summary: Uses graph neural networks to couple detection and temporal association, modelling spatial and temporal relations jointly.

Relation to your work: Valuable as a graph-MOT citation showing why explicit relationships among detections matter. Your paper differs by separating crown detection from post hoc ecological tracking and by allowing split/merge/containment cases instead of only standard MOT identities. ²⁵

```
@inproceedings{wang2020jointgnmot,
  author = {Wang, Yihong and Kitani, Kris and Weng, Xinshuo},
  title = {Joint Object Detection and Multi-Object Tracking with Graph Neural Networks},
  booktitle = {Proceedings of the IEEE International Conference on Robotics and Automation},
  year = {2020}
}
```

RGB-based phenology metrics

1. **Andrew D. Richardson et al., 2009, Ecological Applications.** "Near-Surface Remote Sensing of Spatial and Temporal Variation in Canopy Phenology."

Summary: Foundational paper showing that inexpensive RGB cameras can recover canopy phenology at high temporal frequency.

Relation to your work: Canonical near-surface phenology citation. Your work is best framed as pushing the same logic from fixed phenocam ROIs to moving drone-derived, crown-specific ROIs. ²⁶

```
@article{richardson2009nearsurfacephenology,
  author = {Richardson, Andrew D. and Braswell, Bobby H. and Hollinger, David Y. and Jenkins, Julian P. and Ollinger, Scott V.},
  title = {Near-Surface Remote Sensing of Spatial and Temporal Variation in Canopy Phenology},
  journal = {Ecological Applications},
  year = {2009},
  volume = {19},
  number = {6},
}
```

```

pages = {1417--1428},
doi = {10.1890/08-2022.1}
}

```

1. **Oliver Sonnentag et al., 2012, *Agricultural and Forest Meteorology*.** "Digital Repeat Photography for Phenological Research in Forest Ecosystems."

Summary: Methodological cornerstone for GCC/ExG-style phenology from RGB imagery, including the important result that GCC is generally more robust than ExG to illumination variation.

Relation to your work: Extremely important for your metric section because it directly supports GCC/RCC-style indices and the need to account for illumination effects; your added shadow/texture features are a practical response to those same confounds at crown scale. ²⁷

```

@article{sonnentag2012digitalrepeat,
  author = {Sonnentag, Oliver and Hufkens, Koen and Teshera-Sterne, Cory and others},
  title = {Digital Repeat Photography for Phenological Research in Forest Ecosystems},
  journal = {Agricultural and Forest Meteorology},
  year = {2012},
  volume = {152},
  pages = {159--177},
  doi = {10.1016/j.agrformet.2011.09.009}
}

```

1. **Stephen T. Klosterman et al., 2014, *Biogeosciences*.** "Evaluating Remote Sensing of Deciduous Forest Phenology at Multiple Spatial Scales Using PhenoCam Imagery."

Summary: Shows that PhenoCam-derived phenology metrics can have smaller uncertainty than satellite-derived metrics and discusses fitting/uncertainty issues in event-date estimation.

Relation to your work: Very useful for your future satellite-ground-truth motivation: your crown-level drone signals occupy the scale bridge between near-surface imagery and coarser satellite phenology, exactly the scale mismatch this paper helps motivate. ²⁸

```

@article{klosterman2014evaluating,
  author =
  {Klosterman, Stephen T. and Hufkens, Koen and Gray, Jeffrey M. and others},
  title = {Evaluating Remote Sensing of Deciduous Forest Phenology at Multiple Spatial Scales Using PhenoCam Imagery},
  journal = {Biogeosciences},
  year = {2014},
  volume = {11},
  pages = {4305--4320}
}

```

1. **Jae Yong Park et al., 2019, *Remote Sensing*.** "Quantifying Leaf Phenology of Individual Trees and Species in a Tropical Forest Using Unmanned Aerial Vehicle Images."

Summary: A repeated RGB-UAV study showing that individual crowns and species-level leaf phenology can be retrieved from drone imagery in tropical forest.

Relation to your work: One of your strongest ecological comparators; your paper's novelty over

this type of study is the explicit identity-preserving crown tracking and consensus-ROI construction before phenology extraction. 5

```
@article{park2019leafphenology,  
  author =  
{Park, Jae Yong and Muller-Landau, Helene C. and Lichstein, Jeremy W. and  
others},  
  title = {Quantifying Leaf Phenology of Individual Trees and Species in a  
Tropical Forest Using Unmanned Aerial Vehicle Images},  
  journal = {Remote Sensing},  
  year = {2019},  
  volume = {11},  
  number = {13},  
  pages = {1534},  
  doi = {10.3390/rs11131534}  
}
```

1. **Dominic Fawcett et al., 2021, *Remote Sensing in Ecology and Conservation*.** “Monitoring Spring Phenology of Individual Tree Crowns Using Drone-Acquired NDVI Data.”

Summary: Demonstrates that drone acquisitions can recover crown-scale phenophases and explicitly analyses illumination impacts on drone-derived phenology trajectories.

Relation to your work: Although this study uses multispectral NDVI as the main signal, it is directly relevant because it discusses crown-level phenophase timing and illumination sensitivity, both of which your RGB-only pipeline encounters. 29

```
@article{fawcett2021springphenology,  
  author = {Fawcett, Dominic and Bennie, Jonathan and Anderson, Karen},  
  title = {Monitoring Spring Phenology of Individual Tree Crowns Using Drone-  
Acquired NDVI Data},  
  journal = {Remote Sensing in Ecology and Conservation},  
  year = {2021},  
  volume = {7},  
  number = {2},  
  pages = {227--244},  
  doi = {10.1002/rse2.184}  
}
```

1. **A. D. Richardson et al., 2018, *Scientific Data*; David C. Woebbecke et al., 1995, *Transactions of the ASAE*.** “Tracking Vegetation Phenology Across Diverse North American Biomes Using PhenoCam Imagery” and “Color Indices for Weed Identification Under Various Soil, Residue, and Lighting Conditions.”

Summary: Richardson et al. provide the major modern PhenoCam data resource, while Woebbecke et al. supply the seminal ExG colour-index formulation still used in RGB vegetation analysis.

Relation to your work: These together support the “standard metrics” layer of your phenology feature set: PhenoCam for GCC-style time-series practice and Woebbecke for the long-standing use of RGB greenness indices. 30

```

@article{richardson2018phenocamdata,
  author = {Richardson, Andrew D. and Hufkens, Koen and Milliman, Thomas and others},
  title = {Tracking Vegetation Phenology Across Diverse North American Biomes Using PhenoCam Imagery},
  journal = {Scientific Data},
  year = {2018},
  volume = {5},
  pages = {180028},
  doi = {10.1038/sdata.2018.28}
}

@article{woebbecke1995colorindices,
  author = {Woebbecke, David C. and Meyer, George E. and Von Bargaen, Kenneth and Mortensen, David A.},
  title = {Color Indices for Weed Identification Under Various Soil, Residue, and Lighting Conditions},
  journal = {Transactions of the ASAE},
  year = {1995},
  volume = {38},
  number = {1},
  pages = {259--269}
}

```

Field labels, species mapping, and crown-level ecological interpretation

1. **Geetha Ramaswami, Swati Sidhu, and Suhel Quader, 2021, *Current Science*.** "Using Citizen Science to Build Baseline Data on Tropical Tree Phenology."

Summary: Describes SeasonWatch as an India-wide citizen-science effort for building a baseline dataset on tropical tree phenology across space and time.

Relation to your work: This is your best citation for the Indian citizen-science phenology backdrop. It is especially useful when you discuss why crown-level drone phenology on IIT Delhi can complement broader volunteer-based phenology records in India. ³¹

```

@article{ramaswami2021seasonwatch,
  author = {Ramaswami, Geetha and Sidhu, Swati and Quader, Suhel},
  title = {Using Citizen Science to Build Baseline Data on Tropical Tree Phenology},
  journal = {Current Science},
  year = {2021},
  volume = {121},
  number = {11},
  pages = {1409--1416},
}

```

```
doi = {10.18520/cs/v121/i11/1409-1416}  
}
```

1. **Anastasia Safonova et al., 2021, *Drones*.** "Individual Tree Crown Delineation for the Species Classification and Assessment of Vital Status of Forest Stands from UAV Images."

Summary: Couples individual-tree crown delineation from UAV imagery with species classification and crown-level health/vital-status interpretation.

Relation to your work: This is a strong citation for the ecological value of having per-crown objects rather than only stand-level summaries. Your work differs by making temporal stability and phenological labelling the core output rather than one-off species/vital-status mapping. ³²

```
@article{safonova2021itcdspeciesvitality,  
  author = {Safonova, Anastasia and Tabik, Siham and Peres, Fernando and  
  others},  
  title = {Individual Tree Crown Delineation for the Species Classification  
  and Assessment of Vital Status of Forest Stands from UAV Images},  
  journal = {Drones},  
  year = {2021},  
  volume = {5},  
  number = {3},  
  pages = {77},  
  doi = {10.3390/drones5030077}  
}
```

1. **Yuting Huang et al., 2024, *Remote Sensing*.** "Tree Species Classification from UAV Canopy Images with Deep Learning Models."

Summary: Compares several deep models for tree-species classification from UAV RGB crown images and explicitly studies seasonal transferability.

Relation to your work: Very relevant for the downstream use of your stable crown crops once field species labels are joined. Their weak cross-season transfer result also motivates your choice to separate phenology-state labelling from species classification in a companion paper. ³³

```
@article{huang2024uavspeciesclassification,  
  author = {Huang, Yuting and others},  
  title = {Tree Species Classification from UAV Canopy Images with Deep  
  Learning Models},  
  journal = {Remote Sensing},  
  year = {2024},  
  volume = {16},  
  number = {20},  
  pages = {3836}  
}
```

1. **Sina Soltani et al., 2024, *Biogeosciences*.** "From Simple Labels to Semantic Image Segmentation: Leveraging Citizen Science Plant Photographs for Tree Species Mapping in Drone Imagery."

Summary: Shows that weakly labelled citizen-science photos can be transferred into UAV tree-species segmentation through a two-stage pipeline.

Relation to your work: Strongly relevant as a label-transfer precedent. Their source labels come

from citizen-science photos rather than field forms, but the paper is excellent support for your broader argument that fine-grained drone labels can supervise richer spatial models. ³⁴

```
@article{soltani2024simplelabels,  
  author = {Soltani, Sina and Ferlian, Olga and Eisenhauer, Nico and  
Feilhauer, Hannes and Kattenborn, Teja},  
  title = {From Simple Labels to Semantic Image Segmentation: Leveraging  
Citizen Science Plant Photographs for Tree Species Mapping in Drone Imagery},  
  journal = {Biogeosciences},  
  year = {2024},  
  volume = {21},  
  pages = {2909--2935},  
  doi = {10.5194/bg-21-2909-2024}  
}
```

1. **Giacomo Quattrini et al., 2025, *Remote Sensing*.** “Integrating Drone Truthing and Functional Classification of Remote Sensing Time Series for Supervised Vegetation Mapping.”

Summary: Demonstrates that drone-derived truth data can materially improve supervised classification of Sentinel-2 time-series products by providing large, spatially balanced reference datasets.

Relation to your work: This is the strongest citation for the “drone labels as supervision for coarser satellite models” motivation in your future work section. It is habitat/vegetation mapping rather than crown phenology, but the scale-bridging logic is exactly aligned with your planned companion paper. ³⁵

```
@article{quattrini2025dronetruthing,  
  author = {Quattrini, Giacomo and Pesaresi, Simone and Hofmann, Nicole and  
Mancini, Adriano and Casavecchia, Simona},  
  title =  
{Integrating Drone Truthing and Functional Classification of Remote Sensing  
Time Series for Supervised Vegetation Mapping},  
  journal = {Remote Sensing},  
  year = {2025},  
  volume = {17},  
  number = {2},  
  pages = {330},  
  doi = {10.3390/rs17020330}  
}
```

1. **Tianyi Qin and Qingjian Zhao, 2025, *Scientific Reports*.** “Multi-Branch and Multi-Label Tree Species Classification Using Deep Learning for UAV Aerial Photography and Sentinel Remote Sensing Images.”

Summary: Jointly uses UAV aerial photography and Sentinel data for tree-species classification, making it an unusually direct cross-scale species-mapping paper.

Relation to your work: A strong forward-looking citation for the drone-to-satellite bridge. It is about species rather than phenology, but it supports the exact kind of cross-resolution fusion your project motivates. ³⁶

```
@article{qin2025uavsentinelmultilabel,
  author = {Qin, Tianyi and Zhao, Qingjian},
  title = {Multi-Branch and Multi-Label Tree Species Classification Using
  Deep Learning for UAV Aerial Photography and Sentinel Remote Sensing Images},
  journal = {Scientific Reports},
  year = {2025},
  volume = {15},
  pages = {32710}
}
```

Open questions and limitations

The literature is now quite strong on repeated UAV acquisition, RGB/Mask-R-CNN crown delineation, and near-surface phenology metrics, but it is still comparatively thin on one very specific junction that your paper sits in the middle of: **multi-date crown identity stabilisation from orthomosaic instance masks under residual co-registration error, lighting shifts, partial detections, and split/merge crown events**. The strongest nearby bodies of work are UAV co-alignment papers on one side and graph-based MOT papers on the other, but neither directly solves the crown-consensus problem your method addresses. That gap is worth stating explicitly in your Related Work section because it is, in effect, your paper's clearest novelty claim. ³⁷

A second limitation in the existing literature is that many UAV phenology papers demonstrate that drone imagery can capture crown- or canopy-level dynamics, but far fewer provide a robust, reusable path from those fine-scale labels to coarser satellite phenology models. Quattrini et al. and Qin & Zhao are good bridge citations, yet the specific idea of using **temporally stable drone crown phenology labels** as supervision for satellite phenology remains underdeveloped in the published work you can cite today. That makes your "companion paper" motivation both defensible and timely. ³⁸

¹ ³ UAV-Based Forest Health Monitoring: A Systematic Review

[https://www.researchgate.net/publication/361737476_UAV-Based_Forest_Health_Monitoring_A_Systematic_Review?](https://www.researchgate.net/publication/361737476_UAV-Based_Forest_Health_Monitoring_A_Systematic_Review?utm_source=chatgpt.com)
utm_source=chatgpt.com

² ⁸ Accurate delineation of individual tree crowns in tropical ...

https://zslpublications.onlinelibrary.wiley.com/doi/10.1002/rse2.332?utm_source=chatgpt.com

⁴ (PDF) Monitoring selective logging in a pine dominated forest in Central Germany with repeated drone flights utilizing a RTK quadcopter

[https://www.researchgate.net/publication/348327302_Monitoring_selective_logging_in_a_pine_dominated_forest_in_Central_Germany_with_repeated_drone_flights_utilizing_a_low_cost_RTK_](https://www.researchgate.net/publication/348327302_Monitoring_selective_logging_in_a_pine_dominated_forest_in_Central_Germany_with_repeated_drone_flights_utilizing_a_low_cost_RTK_quadcopter)

⁵ Quantifying Leaf Phenology of Individual Trees and ...

https://www.mdpi.com/2072-4292/11/13/1534?utm_source=chatgpt.com

⁶ Assessment of the Health Status of Old Trees ...

https://www.mdpi.com/2504-446X/8/3/91?utm_source=chatgpt.com

⁷ Urban tree species classification using UAV-based multi ...

https://www.tandfonline.com/doi/full/10.1080/15481603.2021.1974275?utm_source=chatgpt.com

⁹ DeepForest: A Python package for RGB deep learning tree ...

https://besjournals.onlinelibrary.wiley.com/doi/abs/10.1111/2041-210X.13472?utm_source=chatgpt.com

- 10 Individual Tree-Crown Detection in RGB Imagery Using ...
https://www.mdpi.com/2072-4292/11/11/1309?utm_source=chatgpt.com
- 11 Mask R-CNN | IEEE Conference Publication
https://ieeexplore.ieee.org/document/8237584?utm_source=chatgpt.com
- 12 Comparison of Classical Methods and Mask R-CNN for Automatic Tree Detection and Mapping Using UAV Imagery | MDPI
<https://www.mdpi.com/2072-4292/14/2/295>
- 13 A benchmark dataset for canopy crown detection and ... - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC8282040?utm_source=chatgpt.com
- 14 Efficient subpixel image registration algorithms
https://opg.optica.org/ol/fulltext.cfm?uri=ol-33-2-156&utm_source=chatgpt.com
- 15 37 An Image-Based Approach for the Co-Registration of Multi-Temporal UAV Image Datasets | MDPI
<https://www.mdpi.com/2072-4292/8/9/779>
- 16 Improving UAV-SfM time-series accuracy by co-alignment and contributions of ground control or RTK positioning - Utrecht University
<https://research-portal.uu.nl/en/publications/improving-uav-sfm-time-series-accuracy-by-co-alignment-and-contri/>
- 17 Frontiers | Case Report: Optimization of Topographic Change Detection With UAV Structure-From-Motion Photogrammetry Through Survey Co-Alignment
<https://www.frontiersin.org/journals/remote-sensing/articles/10.3389/frsen.2021.626810/full>
- 18 Multi-temporal orthophoto and digital surface model ...
https://www.tandfonline.com/doi/full/10.1080/10106049.2022.2143913?utm_source=chatgpt.com
- 19 Parametric Image Alignment Using Enhanced Correlation ...
https://inria.hal.science/hal-00864385v1/document?utm_source=chatgpt.com
- 20 Graph-Based Data Association in Multiple Object Tracking
https://explore.openaire.eu/search/result?pid=10.1007%2F978-3-031-27818-1_32&utm_source=chatgpt.com
- 21 The Hungarian method for the assignment problem - Kuhn
https://onlinelibrary.wiley.com/doi/abs/10.1002/nav.3800020109?utm_source=chatgpt.com
- 22 Simple Online and Realtime Tracking
https://arxiv.org/abs/1602.00763?utm_source=chatgpt.com
- 23 Simple Online and Realtime Tracking with a Deep Association Metric
https://arxiv.org/abs/1703.07402?utm_source=chatgpt.com
- 24 Graph Neural Based End-to-end Data Association Framework for Online Multiple-Object Tracking
https://arxiv.org/abs/1907.05315?utm_source=chatgpt.com
- 25 Joint Object Detection and Multi-Object Tracking with ...
https://www.ri.cmu.edu/app/uploads/2021/04/camera_ready.pdf?utm_source=chatgpt.com
- 26 Near-surface remote sensing of spatial and temporal variation ...
https://esajournals.onlinelibrary.wiley.com/doi/10.1890/08-2022.1?utm_source=chatgpt.com
- 27 Digital repeat photography for phenological research in forest ...
https://scholars.unh.edu/earthsci_facpub/579/?utm_source=chatgpt.com
- 28 Evaluating remote sensing of deciduous forest phenology ...
https://bg.copernicus.org/articles/11/4305/2014/bg-11-4305-2014.html?utm_source=chatgpt.com

- 29 Monitoring spring phenology of individual tree crowns using ...
https://zslpublications.onlinelibrary.wiley.com/doi/pdf/10.1002/rse2.184?utm_source=chatgpt.com
- 30 Tracking vegetation phenology across diverse North ...
https://www.nature.com/articles/sdata201828?utm_source=chatgpt.com
- 31 SeasonWatch - Tree Phenology Dataset (India)
https://www.gbif.org/dataset/d85d848d-791e-4055-bda5-32c16619ab21?utm_source=chatgpt.com
- 32 Individual Tree Crown Delineation for the Species Classification and Assessment of Vital Status of Forest Stands from UAV Images
<https://www.mdpi.com/2504-446X/5/3/77>
- 33 Tree Species Classification from UAV Canopy Images with Deep Learning Models | MDPI
<https://www.mdpi.com/2072-4292/16/20/3836>
- 34 BG - From simple labels to semantic image segmentation: leveraging citizen science plant photographs for tree species mapping in drone imagery
<https://bg.copernicus.org/articles/21/2909/2024/>
- 35 38 Integrating Drone Truthing and Functional Classification of ...
https://iris.univpm.it/retrieve/9bd45f3d-7324-4455-9936-e736f864aa7a/remotesensing-17-00330.pdf?utm_source=chatgpt.com
- 36 Multi-branch and multi-label tree species classification ...
https://www.nature.com/articles/s41598-025-19827-5?utm_source=chatgpt.com